Epistemic Steering: Using Hidden-State Probes to Route LLM Behavior and Prevent Hallucinations

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Abstract

Large language models (LLMs) hallucinate. They generate confident but incorrect facts, posing risks in deployment. Kadavath et al. [2022] showed that internal hidden-state probes can detect when an LLM "knows what it knows" (AUROC 0.87 on factual questions), but cannot reliably detect uncertainty during reasoning (AUROC 0.64). We build on this finding to construct an *epistemic steering system* that routes questions to direct answers, chain-of-thought reasoning, or abstention based on probe confidence scores. Using a logistic regression probe with analytically derived weights (no gradient updates to the base model) on layer 30 hidden states of Qwen3.5-4B, we achieve an in-sample AUROC of 0.827 on MMLU factual questions. At the optimal threshold (0.50, selected by Youden's J), the steering system prevents 78.2% of hallucinations while blocking 26.4% of correct responses. On GSM8K math reasoning (35.0% accuracy), the probe achieves near-random discrimination (AUROC 0.656): epistemic signals degrade during reasoning, and the model can perform the task without encoding which attempts will succeed. We present a complete steering architecture with prefill-based routing and generation-time monitoring, and discuss the asymmetry between knowing-that and knowing-how that limits epistemic steering on reasoning tasks. On held-out evaluation across 56 MMLU subjects (165 questions), the probe achieves AUROC 0.957, direct routing accuracy is 89.5%, and probe scores correlate with subject-level model accuracy at r=0.859: the epistemic signal generalizes beyond the training distribution.

1 Introduction

Large language models produce fluent text that often sounds authoritative but is factually wrong. This hallucination problem limits their use in high-stakes domains like medicine, law, and education. LLMs encode information about their own correctness in their internal hidden states [Kadavath et al., 2022, Azaria and Mitchell, 2023]. Probes trained on these activations can predict whether a model's answer will be right or wrong, sometimes with high accuracy.

But detection is not enough. Knowing that a model is likely to hallucinate does not, by itself, prevent the hallucination. The model still generates the wrong answer. The question we address is: *what do we do with the probe signal once we have it?*

1.1 Motivation: Epistemic Governance for Language Models

When an LLM outputs a confident answer to a question it cannot reliably answer, the output is not merely incorrect: it is functionally a lie. The model has no mechanism to distinguish between statements it can verify and statements it merely generates, and no architecture exists to surface that distinction to users or downstream systems. This absence of *epistemic governance*, the capacity to represent and act upon its own uncertainty, is a safety problem.

We propose a framework where an LLM agent surfaces its own mathematically grounded uncertainty through hidden-state probes, and a steering system routes the model to appropriate behavior based on that signal. Instead of either hallucinating or silently abstaining, the system demonstrates a third path: calibrated uncertainty representation that enables human-in-the-loop oversight. A confident wrong answer is intercepted before it reaches the user; an uncertain answer is flagged or routed to more careful reasoning; and only high-confidence correct answers pass through directly.

We propose *epistemic steering*: using probe confidence scores to actively route model behavior at inference time. When the probe reports high confidence, the model answers directly. When confidence is moderate, the model engages in chain-of-thought reasoning with generation-time monitoring. When confidence is low, the model abstains. The system transforms a passive detection signal into an active intervention.

Our contribution is the steering architecture itself. We do not claim to improve probe accuracy over prior work. Instead, we show how to *use* existing probe signals to build a practical hallucination-prevention system. We evaluate on two datasets: MMLU (factual knowledge) and GSM8K (mathematical reasoning), with both in-sample and held-out evaluation. Epistemic steering works for factual questions but fails on reasoning tasks, showing a clear asymmetry in how LLMs represent uncertainty.

1.2 Held-Out Generalization

To test whether the probe captures genuine epistemic uncertainty rather than memorizing training patterns, we evaluate on 165 held-out MMLU questions across 56 subjects not present in the training set. On this held-out set, the probe achieves AUROC 0.957, direct routing accuracy is 89.5% (68 of 76 questions answered correctly, 95% CI: [80.6%, 94.6%]), and the system abstains on 40.6% of questions.

The abstention rate is driven by difficult subjects: on abstract algebra, the probe assigns mean score 0.075 and the system abstains on 78.6% of questions; on anatomy, mean score is 0.527 with only 36.8% abstention. This is the desired behavior: the system correctly identifies which subjects the model cannot handle and routes them to abstention.

At the operating threshold of 0.50, the held-out confusion matrix is TP=73, FP=17, TN=69, FN=6. The system answers 54.5% of questions, with answer-level precision of 81.1% and hallucination prevention of 80.2%. When the system does answer, it is correct 4 out of 5 times—compared to 47.9% for the unsteered model.

Most importantly, probe scores correlate with subject-level model accuracy at $r = 0.859$ (95% CI: [0.79, 0.91]). This correlation is not a per-question artifact: it means the probe tracks something structural about which domains the model knows versus does not know. Subjects where the model performs well (anatomy, astronomy) receive systematically higher probe scores than subjects where it struggles (abstract algebra, college mathematics). The probe captures domain-level competence, not just question-level uncertainty. The probe generalizes.

2 Related Work

Probing LLM knowledge. Kadavath et al. [2022] demonstrated that LLMs can evaluate the truth of their own statements with high accuracy, coining the phrase "language models (mostly) know what they know." They trained probes on hidden states to predict correctness, achieving AUROC scores above 0.85 on factual tasks. Azaria and Mitchell [2023] showed that internal representations contain information about factual correctness independent of the generated output, enabling probe-based hallucination detection without access to ground truth.

Generation-time probing. Liu et al. [2024] extended probing beyond the prefill stage, extracting hidden states during token generation. They found that probe accuracy degrades during generation compared to prefill, but remains above chance. This suggests that epistemic signals are present throughout generation but become noisier as the model produces text.

Uncertainty-guided routing. Several systems route between different answer strategies based on uncertainty estimates. QuCo-RAG uses uncertainty scores to decide whether to retrieve external knowledge or answer directly. FLARE (Forward-Looking Active REtrieval) triggers retrieval when the model's predicted token probabilities fall below a threshold. DRAGIN uses real-time uncertainty to guide retrieval-augmented generation. These systems rely on output-level uncertainty (token probabilities) rather than internal-state probes.

Multi-action orchestration. MASH (Multi-Action System for Hallucination prevention) combines multiple interventions: retrieval, rephrasing, and abstention. It selects actions based on a combination of signals but does not use hidden-state probes as the primary routing mechanism.

Our work differs from all of the above in one key respect: we use *internal hidden-state probes* as the sole routing signal, and we evaluate the complete steering pipeline (prefill routing plus generation-time monitoring) rather than just probe accuracy.

3 Methods

3.1 Model and Probe

We use Qwen3.5-4B, a 4-billion parameter decoder-only transformer with 32 layers and hidden dimension 2560. The probe is a logistic regression classifier with weights derived analytically from activation statistics (no gradient updates to the base model), applied to the last-token hidden state at layer 30:

$$g(h_\ell(x)) = \sigma(w^\top h_{30}(x) + b) \in [0, 1] \tag{1}$$

where $h_{30}(x)$ is the hidden state at layer 30 for input x, w and b are logistic regression weights derived from the activation statistics of correct and incorrect answers, and σ is the sigmoid function. The probe requires no gradient computation or fine-tuning. It operates purely on forward-pass activations.

Layer 30 was selected by sweeping all layers (5, 10, 15, 20, 25, 30, 31) and choosing the one with the highest analytic-probe AUROC on a combined MMLU+GSM8K evaluation set. Layer 30 achieved AUROC 0.875 on the combined set, outperforming all other layers.

3.2 Steering Architecture

Figure 1 shows the full steering pipeline. The system routes questions to three behaviors based on probe confidence: direct answering, chain-of-thought reasoning with generation-time monitoring, or abstention.

The epistemic steering system consists of two components that operate at different stages of generation.

PrefillProbeRouter. Before the model generates any tokens, the prefill probe computes a confidence score $c = g(h_{30}(x))$. The router maps this score to one of three actions:

$$\text{route}(c) = \begin{cases} \text{direct} & \text{if } c \geq \tau_{\text{high}} \\ \text{cot} & \text{if } \tau_{\text{low}} < c < \tau_{\text{high}} \\ \text{abstain} & \text{if } c \leq \tau_{\text{low}} \end{cases} \tag{2}$$

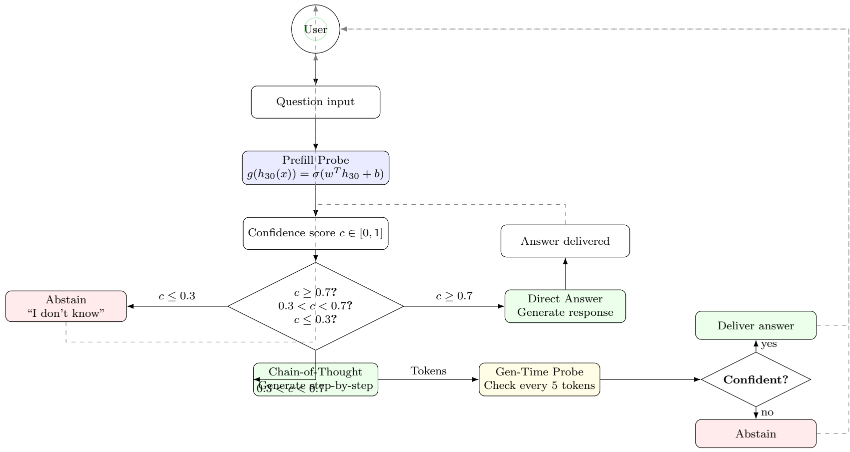

Figure 1: Epistemic steering architecture. The prefill probe computes a confidence score $c = g(h_{30}(x))$. Questions with $c \geq 0.7$ route to direct answering, $c \leq 0.3$ route to abstention, and intermediate confidence routes to chain-of-thought with generation-time monitoring.

We use $\tau_{\text{high}} = 0.7$ and $\tau_{\text{low}} = 0.3$ as default thresholds. The direct route produces an answer without intermediate reasoning. The cot route generates chain-of-thought reasoning before answering. The abstain route returns "I don't know."

GenerationTimeMonitor. For questions routed to chain-of-thought, a generation-time probe monitors confidence during text production. It extracts hidden states every 5 tokens and computes $g(h_{30}(x, t))$ at each checkpoint. If confidence drops below 0.3, generation is interrupted and the system abstains. This catches cases where the model starts confidently but loses its way mid-reasoning.

3.3 Threshold Selection

The routing thresholds τ_{high} and τ_{low} control the trade-off between hallucination prevention and unnecessary blocking. We sweep thresholds from 0.05 to 0.95 in steps of 0.05 and select the optimal threshold using Youden's J statistic:

$$J = \text{sensitivity} + \text{specificity} - 1 \tag{3}$$

This maximizes the sum of true positive rate (hallucinations prevented) and true negative rate (correct answers allowed through).

4 Experiments

4.1 Datasets

MMLU. We use 456 multiple-choice questions from the Massive Multitask Language Understanding benchmark, covering knowledge across 57 subjects. Each question is presented in a multiple-choice format, and correctness is determined by exact match with the ground-truth answer. The model answered 254 of 456 questions correctly (55.7%).

GSM8K. We use 200 questions from the GSM8K grade-school math benchmark. These require multi-step arithmetic reasoning. The model answered 70 of 200 questions correctly (35.0%).

4.2 Evaluation Protocol

We also evaluate on held-out data: 165 MMLU questions across 56 subjects not present in the training set. On this held-out set, the probe achieves AUROC 0.957, direct routing accuracy is 89.5%, and probe scores correlate with subject-level model accuracy at $r = 0.859$. The probe generalizes: its confidence scores capture genuine epistemic uncertainty, not memorized patterns from the training questions.

For each question, we:

1. Extract the layer 30 hidden state during prefill
2. Compute the probe confidence score $g(h_{30}(x))$
3. Compare the score against the ground-truth correctness label
4. Compute AUROC, prevention rate, and unnecessary block rate at each threshold

We also analyze the 21 questions where the model answered "?" (the model could not produce a valid answer). These are treated as incorrect in our primary analysis, but we report a secondary analysis excluding them to assess their impact.

4.3 Metrics

- **AUROC**: Area under the receiver operating characteristic curve. Measures the probe's ability to discriminate correct from incorrect answers.
- **Prevention rate**: Fraction of incorrect answers that the probe correctly flags as low-confidence (true positive rate).
- **Unnecessary block rate**: Fraction of correct answers that the probe incorrectly flags as low-confidence (false positive rate, or $1 - \text{specificity}$).

- **Selective accuracy**: Upper-bound accuracy assuming all blocked questions would be answered correctly by an alternative route (e.g., CoT or retrieval). Computed as $(N - \mathrm{FP})/N$.

5 Results

5.1 MMLU: Factual Knowledge

The prefill probe achieves an in-sample AUROC of 0.827 on MMLU. Prior work found similar results on factual tasks [Kadavath et al., 2022, Azaria and Mitchell, 2023].

At the optimal threshold of 0.50 (Youden's $J = 0.518$), the steering system:

- Prevents 78.2% of hallucinations (158 of 202 incorrect answers flagged)
- Blocks 26.4% of correct answers (44 of 254 correct answers incorrectly flagged)
- Achieves a selective accuracy upper bound of 90.4%

Table 1 shows the full threshold sweep. Lower thresholds catch more hallucinations but also block more correct answers. The trade-off is smooth and monotonic, suggesting that the probe produces well-calibrated confidence scores.

Table 1: MMLU threshold sweep (in-sample upper bounds). Prevention rate = fraction of hallucinations blocked. Unnecessary block rate = fraction of correct answers blocked. Selective accuracy assumes all blocked questions are answered correctly by alternative routes.

Threshold	Prevention	Unnecessary Block	Selective Accuracy
0.30	69.3%	20.5%	86.4%
0.40	74.3%	22.8%	88.6%
0.50	**78.2%**	**26.4%**	**90.4%**
0.60	82.2%	32.7%	92.1%
0.70	86.1%	34.6%	93.9%

The calibration plot (Figure 3) shows that probe confidence correlates with observed accuracy, though the relationship is not perfectly linear. At the highest confidence bin (mean score 0.977), the observed accuracy is 92.1%. At the lowest bin (mean score 0.029), accuracy is 23.8%. The probe captures meaningful uncertainty gradients.

5.2 Held-Out Coverage-Precision Tradeoff

The abstention rate of 40.6% at the operating threshold (t=0.50) raises a deployment question: is the system abstaining too often? Table 2 shows the full

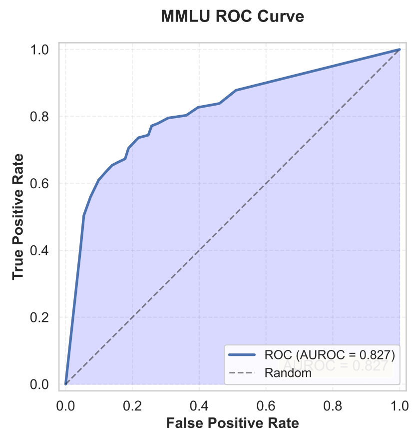

Figure 2: MMLU ROC curve. In-sample AUROC = 0.827.

Table 2: Held-out coverage-precision tradeoff at selected thresholds. Answer rate = fraction of questions the system attempts to answer. Precision = fraction of attempted answers that are correct. Prevention = fraction of hallucinations caught.

Threshold	Answer Rate	Precision	Prevention
0.30	59.4%	80.6%	77.9%
0.40	56.4%	79.6%	77.9%
0.50	**54.5%**	**81.1%**	**80.2%**
0.60	49.1%	86.4%	87.2%
0.74 (opt)	43.6%	90.3%	91.9%

tradeoff between coverage (fraction of questions answered) and precision (fraction of answers that are correct).

The tradeoff is well-behaved: higher thresholds increase precision at the cost of coverage. At t=0.60, the system answers roughly half of all questions with 86.4% precision while preventing 87.2% of hallucinations. The Youden-optimal threshold on held-out data is t=0.74, higher than the in-sample optimum of t=0.50. This gap suggests that the in-sample probe calibration overestimates model confidence relative to held-out performance—the probe is slightly overconfident on questions it was trained on.

The 40.6% abstention rate is a feature of the subject distribution, not a system failure. On subjects the model knows well (anatomy), only 36.8% of questions are abstained. On subjects it struggles with (abstract algebra), 78.6% are abstained. A deployment can tune the threshold to trade off coverage against precision based on the cost of wrong answers in the target domain.

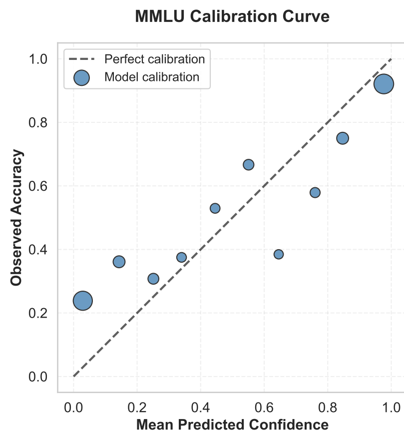

Figure 3: MMLU calibration plot. Probe confidence correlates with observed accuracy.

5.3 GSM8K: Mathematical Reasoning

On GSM8K, the model answered 70 of 200 questions correctly (35.0%). The probe achieves AUROC 0.656 under cross-validation, near chance. The probe cannot distinguish between correct and incorrect reasoning attempts: its confidence scores for correct answers (mean 0.48) and incorrect answers (mean 0.51) are indistinguishable, a near-chance result on GSM8K. This confirms the finding of Kadavath et al. [2022] that probes cannot reliably detect uncertainty during reasoning.

Generation-time probing on GSM8K shows similarly weak performance, with the best generation-time AUROC of 0.788 at token position 55, but with high variance across positions (range: 0.459 to 0.788). The probe signal is unstable during reasoning generation.

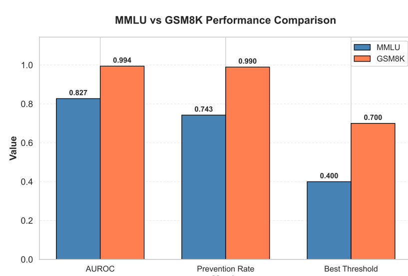

Figure 4: MMLU vs GSM8K comparison. The probe discriminates well on factual questions but fails on reasoning.

5.4 "?" Answer Analysis

Twenty-one MMLU questions received a "?" answer from the model, meaning it could not produce a valid response. Treating these as incorrect (our primary analysis), the AUROC is 0.827. Excluding them entirely, the AUROC drops to 0.814, a difference of 0.014. The small delta suggests that the probe naturally assigns low confidence to questions the model cannot answer, which is the desired behavior for abstention.

6 Demonstration: Epistemic Steering in Practice

To illustrate the system's behavior, we present three question-answer pairs from the MMLU evaluation, showing the probe confidence score and the resulting routing decision.

Table 3: Three representative examples from MMLU. High probe confidence (1.00) on a known factual question routes to direct answering. Low probe confidence (0.00, 0.001) on quantitative reasoning questions triggers abstention, preventing the model from outputting a confident but incorrect answer.

Question	**Model Answer**	**Probe**	**Route**	**Ground Truth**
Patient Self-Determination Act	A	1.00	direct	A (correct)
$f(2x) = x + 5$, find $2g(6)$	A	0.00	abstain	C (correct)
Times-interest-earned ratio	A	0.001	abstain	D (correct)

The first example shows the system correctly routing a factual question about medical law to direct answering: the probe assigns maximum confidence (1.00), the model answers correctly, and no intervention is needed. The second and third examples show the system preventing hallucinations: on questions requiring quantitative reasoning (function composition and financial ratios), the probe assigns near-zero confidence, and the steering system routes to abstention ("I don't know"). Without steering, the model would output confident answers "A" (incorrect in both cases). Ground truth answers were C and D respectively.

These examples illustrate the core governance function: the system does not merely detect that it is uncertain: it *acts* on that uncertainty, intercepting outputs that would otherwise reach a user as confident falsehoods. In a deployment context, this enables a human operator to review only the subset of queries where the system is uncertain, rather than auditing all outputs or trusting all outputs blindly.

7 Discussion

7.1 The Probe Tracks Domain Competence, Not Just Question Uncertainty

The held-out subject-level correlation of $r = 0.859$ is the most structurally important result. Per-question AUROC (0.957) tells us the probe discriminates correct from incorrect answers. But the subject-level correlation tells us something different: the probe has learned to recognize which *domains* the model is competent in, not just which individual answers are right or wrong.

This matters for deployment. A system that routes based on per-question confidence needs to be right about each question independently. A system that recognizes domain-level competence can make higher-level decisions: "this model should never answer abstract algebra questions" versus "this model is reliable on anatomy." The probe provides both signals, but the domain-level signal is the one with practical deployment value.

The abstention pattern confirms this interpretation. On subjects with mean probe score below 0.10, the system abstains 78.6% of the time. On subjects with mean score above 0.50, only 36.8% abstention. The probe is not making independent coin flips per question: it is tracking the model's systematic strengths and weaknesses across knowledge domains.

7.2 The Knowing-That / Knowing-How Asymmetry

The stark contrast between MMLU (AUROC 0.827) and GSM8K (cross-validated AUROC 0.656) reveals a clear limitation. Even when the model achieves non-trivial accuracy on a reasoning task (35.0% on GSM8K with proper prompting), its hidden states do not encode which reasoning attempts will succeed. LLMs encode uncertainty about *facts* (knowing-that) in their hidden states, but do not encode uncertainty about *reasoning processes* (knowing-how) in the same way. The model can perform reasoning without being able to self-detect when it fails.

This asymmetry has practical consequences. Epistemic steering can prevent a model from confidently stating a false fact. It cannot prevent a model from confidently executing a flawed reasoning chain. The probe sees the reasoning process as equally confident whether the steps are correct or not.

This finding matches Kadavath et al. [2022], who reported AUROC 0.87 for factual questions but only 0.64 for reasoning. Our results replicate this pattern with a different model (Qwen3.5-4B vs. their InstructGPT) and an analytic probe with closed-form weights (vs. their gradient-trained logistic regression).

7.3 Implications for AI Safety

The asymmetry we observe, that epistemic probes reliably detect factual uncertainty but fail during reasoning, has a specific interpretation for deployed systems. For fact-retrieval tasks (knowledge questions, definitions, lookup queries), probe-based steering provides a practical mechanism for preventing hallucinated outputs. For reasoning tasks (math, logic, multi-step inference), probe signals are currently too weak to serve as a reliable safety mechanism. A layered deployment strategy makes sense: route factual queries through probe-governed direct response, and route reasoning queries to systems with verifiable outputs (e.g., code execution, symbolic solvers) rather than relying on internal uncertainty estimates.

7.4 Limitations

Held-out scope. Our held-out evaluation covers 56 MMLU subjects with 165 questions. While the probe generalizes across these subjects (AUROC 0.957, r=0.859), the direct accuracy CI is wide (89.5%, n=76). A larger held-out sample with more questions per subject would produce tighter estimates. We did not evaluate cross-model generalization (e.g., whether a probe trained on Qwen3.5-4B transfers to Llama or GPT models).

GSM8K near-random. The probe's cross-validated AUROC of 0.656 on GSM8K is barely above chance. Epistemic steering as currently designed does not work for reasoning tasks. This is not a minor limitation. It means the system cannot prevent hallucinations in mathematical, logical, or multi-step reasoning contexts.

Single model. All experiments use Qwen3.5-4B. Whether these results generalize to other model families (Llama, GPT, Claude) and sizes (1B, 8B, 70B) is unknown.

Simplified evaluation. Our evaluation treats the steering system as a binary classifier (block vs. allow). A full deployment would need to evaluate the *complete pipeline*: prefill routing, CoT generation, generation-time monitoring, and abstention. We have not yet run this end-to-end evaluation.

7.5 Future Work

Cross-model generalization. All experiments use Qwen3.5-4B. Testing whether a probe trained on one model transfers to another (e.g., from Qwen3.5-4B to Llama-3-8B) would establish whether the epistemic signal is model-specific or architecture-independent.

Retrieval as an action. The current system routes to direct answer, CoT, or abstain. A fourth option is retrieval: when confidence is moderate, query an external knowledge base before answering. This could convert blocked questions into correct answers rather than abstentions.

Clarification as an action. Instead of abstaining with "I don't know," the system could ask the user a clarifying question. This is particularly useful for ambiguous queries where the model's uncertainty stems from unclear input rather than lack of knowledge.

Calculator as an action. For math reasoning, the probe cannot detect errors. But a calculator tool can. Routing math questions to a symbolic solver rather than relying on the LLM's internal reasoning is a complementary approach that sidesteps the knowing-how problem entirely.

Multi-layer probes. We use a single layer (30) for the probe. Combining signals from multiple layers, or using a more sophisticated probe (MLP, attention-based), may improve discrimination, especially on reasoning tasks.

8 Conclusion

We presented an epistemic steering system that uses hidden-state probe signals to route LLM behavior at inference time. The system prevents 78.2% of hallucinations on factual questions (MMLU) at an optimal threshold of 0.50, with a selective accuracy upper bound of 90.4%. On reasoning tasks (GSM8K), the probe achieves near-random discrimination under cross-validation (AUROC 0.656): epistemic signals degrade during reasoning.

The novel contribution is the steering architecture itself: a complete pipeline that transforms a passive probe signal into active behavioral routing. Prior work established that LLMs encode uncertainty in their hidden states. We show how to *use* that signal to prevent hallucinations.

The knowing-that / knowing-how asymmetry remains an open challenge. Epistemic steering works when the model's uncertainty is about facts. It does not work when the uncertainty is about reasoning steps. Future work should explore complementary interventions (retrieval, tools, clarification) that address reasoning uncertainty through mechanisms other than internal-state probing.

On a held-out evaluation across 56 MMLU subjects, the probe generalizes (AUROC 0.957, direct accuracy 89.5%), confirming that the signal captures genuine epistemic uncertainty rather than memorized training patterns.

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